Homework 7: Linear Transformations and Projections

EECS 245, Spring 2026 at the University of Michigan

due Thursday, June 4th, 2026 at 11:59PM Ann Arbor Time

Write your solutions to the following problems either by writing them on a piece of paper or on a tablet and scanning your answers as a PDF. Note that you are not allowed to use LaTeX, Google Docs, or any other digital document creation software to type your answers. Homeworks are due to Gradescope by 11:59PM on the due date. See the **syllabus** for details on the slip day policy.

Homework will be evaluated not only on the correctness of your answers, but on your ability to present your ideas clearly and logically. You should always explain and justify your conclusions, using sound reasoning. Your goal should be to convince the reader of your assertions. If a question does not require explanation, it will be explicitly stated.

Before proceeding, make sure you're familiar with the **collaboration policy**.

Total Points: 10 + 35 + 10 + 8 = 63

Problem 1: Homework 6 Solutions Review (10 pts)

Review the solutions to Homework 6. Pick **two problem parts** (for example, Problem 2a and Problem 5b) from Homework 6 in which your solutions have the most room for improvement, i.e., where they have unsound reasoning, could be significantly more efficient or clearer, etc. **Include a screenshot of your solution to each problem part**, and in a few sentences, explain what was deficient and how it could be fixed.

Alternatively, if you think one of your solutions is significantly better than the posted one, copy it here and explain why you think it is better. If you didn't do Homework 6, choose two problem parts from it that look challenging to you, and in a few sentences, explain the key ideas behind their solutions in your own words.

Problem 2: Projecting onto the Column Space (35 pts)

The big idea in **Chapter 6.3**, which we will also revisit in Chapter 7.1, is that of projecting a vector $\vec{y} \in \mathbb{R}^n$ onto the column space of an $n \times d$ matrix X. That is, unlike in previous homeworks, we **don't** assume $\vec{y}$ is a linear combination of X's columns, and instead aim to find the vector in $\text{colsp}(X)$ that is closest to $\vec{y}$.

As we will show in **Chapter 6.3**, if X's columns are the vectors $\vec{x}^{(1)}, \vec{x}^{(2)}, \ldots, \vec{x}^{(d)}$, then the projection of $\vec{y}$ onto $\text{colsp}(X)$ is the vector

$$\vec{p} = X\vec{w}^* = w_1^*\vec{x}^{(1)} + w_2^*\vec{x}^{(2)} + \cdots + w_d^*\vec{x}^{(d)}, \quad \text{where} \quad \underbrace{\vec{w}^* = (X^TX)^{-1}X^T\vec{y}}_{\text{if } X^TX \text{ is invertible}}$$

The vector $\vec{w}^*$ minimizes the norm of the error vector, $\vec{e} = \vec{y} - \vec{p}$, and as we will see in coming problems and homeworks, it contains **optimal model parameters** for linear regression, when we fill our X (carefully) with our input variables and $\vec{y}$ with our output variables.

Taking another look at the formula $\vec{p} = X\vec{w}^*$, we see that it's equivalent to

$$\vec{p} = X\vec{w}^* = X(X^TX)^{-1}X^T\vec{y} = P\vec{y}$$

where $P = X(X^TX)^{-1}X^T$ is called the **projection matrix**. Multiplying $P\vec{y}$ is equivalent to projecting $\vec{y}$ onto $\text{colsp}(X)$. In this problem, you'll explore properties of P and $\vec{w}^*$.

a) (4 pts) In this part only, suppose X is an $n \times 1$ matrix, i.e. it is a vector. Then,

(i) What is the value of $\vec{w}^*$, and how does it relate to what we learned in **Chapter 3.4**?

(ii) What is the value of the matrix P, and how does it relate to what we learned in **Homework 6, Problem 5**?

b) (4 pts) Show that P is both symmetric (meaning that $P^T = P$) and idempotent (meaning that $P^2 = P$). Then, explain in English how P's idempotence relates to the linear transformation of projecting $\vec{y}$ onto colsp(X).

c) (2 pts) Recall that X is an $n \times d$ matrix (meaning it's not necessarily square), which makes $P = X(X^TX)^{-1}X^T$ an $n \times n$ matrix.

Fill in the blanks: X^TX is invertible if and only if X's columns are ____.

d) (2 pts) In the rare case that X is an $n \times n$ square matrix, and rank$(X) = n$, what is P? What does this say about the relationship between $\vec{y}$, $\vec{p}$, and colsp(X)?

Hint: Use the fact that $(AB)^{-1} = B^{-1}A^{-1}$.

e) (8 pts) In **(i)** and **(ii)**, find $\vec{w}^*$, $\vec{p}$, and $\vec{e} = \vec{y} - \vec{p}$, and verify that $\vec{e}$ is orthogonal to colsp(X) by showing that it is orthogonal to each of X's columns.

(i) $X = \begin{bmatrix} 2 & 1 \\ 0 & -3 \\ 0 & 0 \end{bmatrix}, \quad \vec{y} = \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix}$

(ii) $X = \begin{bmatrix} 1 & 1 \\ 1 & -1 \\ 1 & 0 \end{bmatrix}, \quad \vec{y} = \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix}$

(iii) In **only one** of the subparts above, it is true that the sum of the components of $\vec{e}$ is 0. Which one, and why? This is a **hugely** important result, and one that will 100% appear on Midterm 2. *Hint: The answer is not that $\vec{y}$ is in colsp(X); in both parts, it is **not**.*

f) (2 pts) What linear combination of $\begin{bmatrix} 1 \\ 2 \\ -1 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ 0 \\ 1 \\ -3 \end{bmatrix}$ is closest to $\begin{bmatrix} 3 \\ 1 \\ 2 \\ 4 \end{bmatrix}$?

g) (3 pts) Find the point on the plane $x - y - 2z = 0$ that is closest to the point $(5, 0, 3)$.

Hint: Start by thinking of this as a projection problem. What are X and $\vec{y}$?

h) (6 pts) Consider the matrix

$$A = \begin{bmatrix} 3 & 6 & 6 \\ 4 & 8 & 8 \end{bmatrix}$$

(i) How many vectors $\vec{w}$ are there that minimize $\|\vec{y} - A\vec{w}\|^2$? 0, 1, finitely many, or infinitely many?

(ii) Find the matrix P_c that projects vectors in $\mathbb{R}^2$ onto the column space of A. *Hint: The formula for P will fail here, since $A^T A$ will not be invertible. Start by explaining why $A^T A$ is not invertible. Then, look at Homework 5 for inspiration.*

(iii) Find the matrix P_r that projects vectors in $\mathbb{R}^3$ onto the row space of A.

(iv) Find the product $P_c A P_r$ and explain the result.

i) (4 pts) Suppose that Q is an $n \times d$ matrix whose columns are orthonormal, meaning that its columns are orthogonal to each other and have length 1. Note that we are not assuming that Q's rows are orthonormal too, meaning that Q is not necessarily an orthogonal matrix using the definition from Homework 5 or Chapter 6.1.

The fact that Q's columns are orthonormal greatly simplifies the formulas involved in projecting $\vec{y}$ onto colsp(Q).

(i) What is the value of $\vec{w}^*$?

(ii) What is the value of $\vec{p}$?

(iii) If Q's rows were orthonormal too, how would the answers to **(i)** and **(ii)** simply further, and why?

Problem 3: Moving Things Around (10 pts)

Let X be an $n \times 4$ design matrix whose first column is all 1s, let $\vec{y}$ be an observation vector, and let $\vec{w}^* = (X^T X)^{-1} X^T \vec{y}$, where $\vec{w}^* = \begin{bmatrix} w_0^* \\ w_1^* \\ w_2^* \\ w_3^* \end{bmatrix}$.

In this problem, you'll reason about modifications to the design matrix and see how they affect the components of $\vec{w}^*$.

a) (3 pts) Let X_a be the design matrix that results from **swapping the first two columns of** X. Let $\vec{v}^* = (X_a^T X_a)^{-1} X_a^T \vec{y}$. Express the components of $\vec{v}^*$ in terms of $w_0^*, w_1^*, w_2^*, w_3^*$.

b) (3 pts) Let X_b be the design matrix that results from **adding 3 to each entry in the *first* column of** X. Let $\vec{v}^* = (X_b^T X_b)^{-1} X_b^T \vec{y}$. Express the components of $\vec{v}^*$ in terms of $w_0^*, w_1^*, w_2^*, w_3^*$.

c) (4 pts) Let X_c be the design matrix that results from **adding 3 to each entry in the *second* column of** X. Let $\vec{v}^* = (X_c^T X_c)^{-1} X_c^T \vec{y}$. Express the components of $\vec{v}^*$ in terms of $w_0^*, w_1^*, w_2^*, w_3^*$.

Problem 4: The Sum of Errors (8 pts)

Consider a set of n points, $(\vec{x}_1, y_1), (\vec{x}_2, y_2), ..., (\vec{x}_n, y_n)$, where each $\vec{x}_i$ is a feature vector in $\mathbb{R}^d$ and each y_i is a scalar.

a) (4 pts) To fit the model

$$h(\vec{x}_i) = w_0 + w_1 x_i^{(1)} + w_2 x_i^{(2)} + ... + w_d x_i^{(d)} = \vec{w} \cdot \text{Aug}(\vec{x}_i)$$

we minimize mean squared error,

$$R(\vec{w}) = \frac{1}{n}\sum_{i=1}^{n}(y_i - \vec{w} \cdot \text{Aug}(\vec{x}_i))^2 = \frac{1}{n}\|\vec{y} - X\vec{w}\|^2$$

meaning that $\vec{w}^*$ is chosen to satisfy the normal equations. Explain why the components of the error vector,

$$\vec{e} = \vec{y} - X\vec{w}^*$$

are **guaranteed** to sum to 0.

b) (4 pts) If we decide instead to fit the model

$$h(\vec{x}_i) = w_1 x_i^{(1)} + w_2 x_i^{(2)} + ... + w_d x_i^{(d)} = \vec{w} \cdot \vec{x}_i$$

which has no intercept term, are the components of the error vector $\vec{e} = \vec{y} - X\vec{w}^*$ still guaranteed to sum to 0? If they are, explain why. If they are not, explain why not, but give at least one example dataset where they still do sum to 0.